APMA 3150

# HOW WELL CAN TEMPORAL, SPATIAL, AND METEOROLOGICAL FACTORS PREDICT U.S. ACCIDENTS?


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## BACKGROUND

**Why do some crashes cause major injuries while others do not?**

Understanding crash severity is critical to preventing serious outcomes on U.S. roadways. Prior work has shown that environmental factors like visibility and precipitation play a role, but few studies have unpacked how these conditions interact with time, place, and infrastructure in a rigorous, interpretable way.

## RESEARCH GOAL

**To what extent can weather, lighting, timing, and infrastructure conditions predict crash severity?**

To answer this, we used a multifaceted approach:

- **One-way and Two-way ANOVA** to test how individual and interacting conditions (e.g., rain at night) affect severity.
- **Principal Component Analysis (PCA)** to identify latent environmental or road-related risk profiles.
- **K-Means Clustering** to categorize common high-risk scenarios, such as low-visibility nights or cold sunrises.
- **Logistic Regression** for interpretable baseline modeling of severe vs. non-severe outcomes.
- **Random Forests** to capture non-linear interactions and rank feature importance.
- **Hypothesis Tests and Confidence Intervals** to support exploratory findings with formal statistical inference.

## US ACCIDENTS DATA

**To investigate this, we used the US Accidents (2016–2023) dataset**

- One of the largest open-access traffic crash datasets, featuring over **7 million unique crash records.** It was compiled from real-time streams (MapQuest and Bing Maps APIs) and enriched with data from nearly **2,000 weather stations** and OpenStreetMap road features.

**Each record includes up to 45 attributes**

- Such as GPS location, timestamp, road features (e.g., bumps, traffic signals), weather conditions (temperature, humidity, wind speed, precipitation), and a **severity score from 1 to 4.**

**Prior applications of this dataset include:**

- **Moosavi et al. (2019):** Developed deep learning models to forecast crashes within 30 minutes using this dataset; achieved **AUROC = 0.88**, but lacked interpretability
- **Yan & Shen (2022):** Applied random forests to predict severity **(accuracy: 76–80%)**, identifying visibility and wind speed as key predictors
- **Nippani et al. (2024):** Used graph neural networks to model road-based accident likelihood; high performance but limited transparency

## WHY CRASHES GET SERIOUS

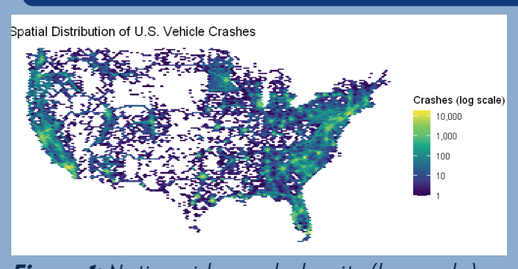



***Figure 1:*** *Nationwide crash density (log scale)*

Crash reports cluster along the **eastern seaboard,** the **Los-Angeles basin**, and the **Texas–Illinois corridor**. While volume is unsurprisingly highest in dense metros, later analyses show that **sheer frequency alone is a poor predictor of severe outcomes.**

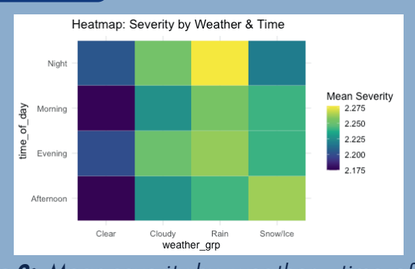



***Figure 2:*** *Mean severity by weather × time-of-day*

The two-way ANOVA pinpoints the riskiest windows: **nighttime rain and dawn snow/ice increase** the four level severity score by as much as **0.16** relative to clear daytime driving, underscoring how limited visibility compounds weather hazards.

## DRIVERS AND INTERVENTIONS

A **200-tree random forest**, trained on **one million accidents** and evaluated on a **30 % hold-out set**, explains **37% of the out-of-sample variation** in the **four-level severity score**—six times better than an OLS baseline. Four weather fields **(wind-chill, air temperature, humidity, wind speed)** deliver nearly half that lift; location variables lag far behind. In short, what the driver feels through the windshield outweighs where the driver is on the map

The ANOVA then points to a cheap **counter-measure**: on wet pavement **a simple speed bump offsets roughly two-thirds of the visibility penalty** and costs a rounding error next to full lighting upgrades. Diagnostics confirm the model is trustworthy—**residuals are near-normal with constant spread**—so the bump result is robust

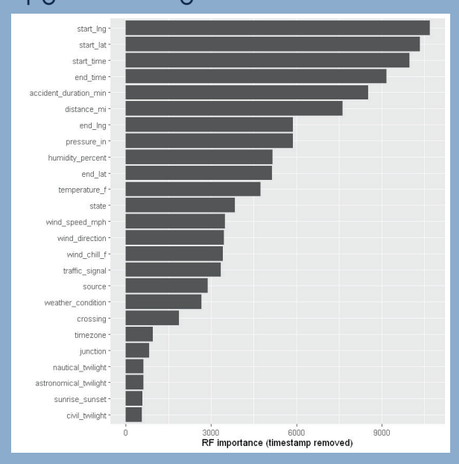



**Figure 3:** Key drivers. Weather dominates the importance chart; latitude and longitude barely register.

Switching from OLS to the random forest raises hold-out R 2 from 0.06 to 0.37. The top four slots all belong to micro-weather, confirming that **air, not address, predicts injury.**

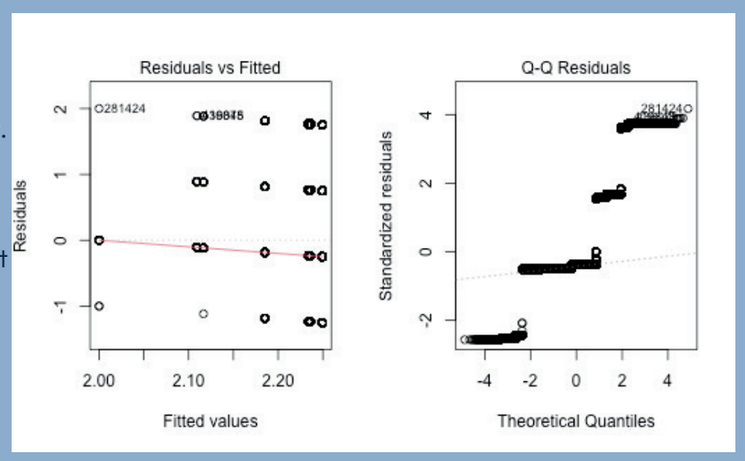



**Figure 4:** Assumption checks. Residual-vs-fitted (left) and Q-Q (right) show no funnel and no heavy tails.

With normal, homoscedastic residuals, the F-tests and Tukey contrasts highlighting the bump benefit estimates are statistically sound.

**Takeaway:**
**Weather** raises **risk**; **bumps blunt** it.
Marrying micro-climate data with low-cost road hardware offers a pragmatic path to fewer severe crashes.

## CONCLUSION

**We conclude that crash severity spikes when low visibility weather overlaps with already difficult driving contexts.**

- **Weather is a major factor: Rain or snow** increases average **severity** by **0.07–0.10 points**, with the **wet/dry flag showing the largest shift (0.10)**.
- **Atmospheric variables** (visibility, humidity, wind chill, etc.) explain more variation in severity than infrastructure features.
- **Nighttime intersections** were more dangerous, showing the role of visibility again
- **Weekend storms** were more severe than weekday ones, possibly due to riskier driving behavior
- **Speed** bumps helped: They **reduced severity by 0.05–0.07** points, especially in rain

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1. National Highway Traffic Safety Administration. (2023, April). *Traffic safety facts: Crash stats, early estimate of motor vehicle traffic fatalities in 2022* (Crash Stats Brief Statistical Summary DOT HS 813 428). U.S. Department of Transportation.
2. Moosavi, S., Samavatian, M. H., Parthasarathy, S., Teodorescu, R., & Ramnath, R. (2019). Accident risk prediction based on heterogeneous sparse data: New dataset and insights. *In Proceedings of the 27th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS 2019)* (pp. 33-42). ACM.
3. Moosavi, S., Samavatian, M. H., Parthasarathy, S., & Ramnath, R. (2019). A countrywide traffic accident dataset. *CoRR*, abs/1906.05409. https://arxiv.org/abs/1906.05409
4. Nippani, A., Li, D., Ju, H., Koutsopoulos, H. N., & Zhang, H. R. (2024). *Graph neural networks for road safety modeling: Datasets and evaluations for accident analysis.* https://arxiv.org/abs/2311.00164